EXISTENCE OF THE DERIVED PULLBACK AGREEING WITH THE NAIVE PULLBACK ON K-FLAT COMPLEXES WITH FLAT TERMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete verification: the hypotheses of the abstract derived-functor result are checked in this situation and the identification of the values is attributed to the corresponding result of this chapter. The whole verification lies inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Derived tensor product and K-flat complexes on a ringed topos. Same chapter, the derived tensor product and the role of K-flat representatives in computing it. Supplies the machinery through which the functor Lf^* is constructed.
- General theory of the derived functor of an additive functor between triangulated categories. Derived Categories, the construction of a left derived functor through a suitable class of complexes and the verification that it is exact and triangulated. Supplies the abstract result which the author applies to the pullback functor f^*.
- Pullback of K-flat complexes with flat terms. Same chapter, the result that the pullback of a K-flat complex with flat terms is again K-flat with flat terms. Supplies the identification Lf^*K^bullet = f^*K^bullet for the complexes mentioned in the statement, and the bounded above case through the flatness of the terms.

Source reference. Original lemma 05S7: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3599–3599. Navigation pointer invoked at proof

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line 3599; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05S8: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3607–3607. Navigation pointer invoked at proof line 3607; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05S9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3610–3610. Navigation pointer invoked at proof line 3610; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06XN: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3613–3613. Navigation pointer invoked at proof line 3613; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06YS: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3207–3207. Navigation pointer invoked at proof line 3616; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0G7E: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3508–3508. Navigation pointer invoked at proof line 3632; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06YT: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3233–3233. Navigation pointer invoked at proof line 3636; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05SW: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3646–3646. Navigation pointer invoked at proof line 3646; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 06YQ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3120–3120. Navigation pointer invoked at proof line 3539; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}) \to (Sh(\mathcal{C}'), \mathcal{O}')$ be a morphism of ringed topoi. There exists an exact functor*

$$Lf^* : D(\mathcal{O}') \longrightarrow D(\mathcal{O})$$

of triangulated categories so that $Lf^*\mathcal{K}^\bullet = f^*\mathcal{K}^\bullet$ for any K-flat complex $\mathcal{K}^\bullet$ with flat terms and in particular for any bounded above complex of flat $\mathcal{O}'$-modules.

Proof. To see this we use the general theory developed in Derived Categories, Section 05S7. Set $\mathcal{D} = K(\mathcal{O}')$ and $\mathcal{D}' = D(\mathcal{O})$. Let us write $F : \mathcal{D} \to \mathcal{D}'$ the exact functor of triangulated categories defined by the rule $F(\mathcal{G}^\bullet) = f^*\mathcal{G}^\bullet$. We let S be the set of quasi-isomorphisms in $\mathcal{D} = K(\mathcal{O}')$. This gives a situation as in Derived Categories, Situation 05S8 so that Derived Categories, Definition 05S9 applies. We claim that LF is everywhere defined. This follows from Derived Categories, Lemma 06XN with $\mathcal{P} \subset \mathrm{Ob}(\mathcal{D})$ the collection of K-flat complexes $\mathcal{K}^\bullet$ with flat terms. Namely, (1) follows from Lemma 06YS and to see (2) we have to show that for a quasi-isomorphism $\mathcal{K}_1^\bullet \to \mathcal{K}_2^\bullet$ between elements of $\mathcal{P}$ the map $f^*\mathcal{K}_1^\bullet \to f^*\mathcal{K}_2^\bullet$ is a quasi-isomorphism. To see this write this as

$$f^{-1}\mathcal{K}_1^\bullet \otimes_{f^{-1}\mathcal{O}'} \mathcal{O} \longrightarrow f^{-1}\mathcal{K}_2^\bullet \otimes_{f^{-1}\mathcal{O}'} \mathcal{O}$$

The functor f^{-1} is exact, hence the map $f^{-1}\mathcal{K}_1^\bullet \to f^{-1}\mathcal{K}_2^\bullet$ is a quasi-isomorphism. The complexes $f^{-1}\mathcal{K}_1^\bullet$ and $f^{-1}\mathcal{K}_2^\bullet$ are K-flat complexes of $f^{-1}\mathcal{O}'$-modules by Lemma 0G7E because we can consider the morphism of ringed topoi $(Sh(\mathcal{C}), f^{-1}\mathcal{O}') \to (Sh(\mathcal{C}'), \mathcal{O}')$. Hence Lemma 06YT guarantees that the displayed map is a quasi-isomorphism. Thus we obtain a derived functor

$$LF : D(\mathcal{O}') = S^{-1}\mathcal{D} \longrightarrow \mathcal{D}' = D(\mathcal{O})$$

see Derived Categories, Equation (05SW). Finally, Derived Categories, Lemma 06XN also guarantees that $LF(\mathcal{K}^\bullet) = F(\mathcal{K}^\bullet) = f^*\mathcal{K}^\bullet$ when $\mathcal{K}^\bullet$ is in $\mathcal{P}$. The proof is finished by observing that bounded above complexes of flat modules are in $\mathcal{P}$ by Lemma 06YQ. □